

BFS – Breadth First Search:

Definition:

BFS is a graph/tree traversal algorithm that visits nodes level by level. It uses a Queue (FIFO).

Algorithm:

1. Start from the given node.
2. Mark the node as visited.
3. Insert it into the queue.
4. Remove a node from the queue.
5. Visit all its unvisited adjacent nodes.
6. Add those nodes to the queue.
7. Repeat until the queue is empty.

Out put:

```
Enter number of vertices: 5
Enter number of edges: 5
Enter edges (u v):
0 1
0 2
1 3
1 4
2 4
Enter starting vertex: 0
BFS Traversal: [0, 1, 2, 3, 4]
Execution Time: 1.9381055608391762e-05 seconds
Time Complexity: O(V + E)

...Program finished with exit code 0
Press ENTER to exit console.
```

Time Complexity: $O(V + E)$

Space Complexity: $O(V)$